IE 306 — Assignment 3: Drone Light Show Depot — Output Analysis

Technical Report | Spring 2026 | Due: Monday, June 1, 2026

---

1. System Description

The Bosphorus Drone Light Festival operates a **closed-loop fleet of N = 200 drones** cycling continuously between an aerial light show and a ground depot. Each drone returns when its battery falls to ≤ 15 % and passes through:

1. **Swap queue** → one of N_swap identical swap stations (FCFS)
2. **Test rig** → a single shared airworthiness rig
3. **Re-launch** with a full battery pack

Service times are **lognormal** throughout:

Process	Mean (s)	CV
Battery swap	45.0	0.30
Airworthiness test	8.0	0.25
Battery drain multiplier	1.0	0.18

Two configurations are compared:

Policy	Swap stations	Test rigs
A — Current depot	5	1
B — Proposed upgrade	6	1

The simulator is implemented in SimPy (`model.py`) with CRN-aware random streams (`seeds.py`). All parameters are defined in `config.py`. **None of these files were modified.**

---

2. Task 1 — System Classification

Classification: Steady-State

The fleet operates in a **continuous closed loop** with no natural termination event — rehearsal blocks run indefinitely. We are interested in long-run average performance (mean swap-queue wait) rather than behaviour over a fixed episode. The initial condition — all 200 drones launched simultaneously with full batteries — creates an artificial transient (the first wave of returns clusters near $t \approx 2\,400$ s) that is not representative of normal operations. Since we want to evaluate swap station capacity under the *stationary* operating regime, a steady-state analysis is the appropriate framework. Accordingly, a warmup deletion period is applied before computing statistics.

---

3. Task 2 — Warmup Detection

3.1 Welch's Method (Graphical, R = 10 Replications)

Ten independent replications (12 h each, Policy A) were run. Swap-queue wait observations were binned into 60 s intervals; the cross-replication mean was smoothed with a ±60-bin (±60 min) movingaverage window. The smoothed curve was plotted and stabilised around **8 820 s (≈ 2.45 h)**, detected as the first bin where `|smoothed − final_mean| < 0.10 × final_mean`.

warmup_welch = 8 820 s

3.2 MSER (Algorithmic, Single Long Run)

A single 30 h long run (Policy A) minimised the MSER statistic var(x_{L:n}) / (n − L) over the per-completion observation series.

warmup_mser = 3 329 s

3.3 Discrepancy Investigation

The two estimates differ by a factor of **≈ 2.65**, exceeding the 25 % guideline.

Root cause: MSER operates on the *completion-index* axis (not wall-clock time). All 200 drones launch simultaneously at $t = 0$ with full batteries, so the first depot completions only begin around $t \approx 2\,000$–2 400 s. The early observations are sparse in wall-clock time but appear as a brief, high-variance prefix in observation-index space — MSER consequently truncates prematurely. Welch's method averages on a wall-clock grid across multiple replications, better capturing the clustered first-wave burst, and yields a more conservative and reliable estimate.

We adopt Welch as the primary reference.

3.4 Chosen Warmup

`warmup_chosen = 10 000 s  (≈ 2.78 h)`

Exceeds `warmup_welch` (8 820 s) by ~13 % for conservatism.

---

4. Task 3 — Two Confidence Intervals for Policy A

4.1 Method (a): Replications-with-Deletion

R = 40 replications (12 h each, rep_index 0–39). Warmup of 10 000 s deleted from each. One replication mean computed per run; two-sided t-CI from those R means.

Parameter	Value
R (replications)	40
Warmup deleted	10 000 s
Point estimate	**14.53 s**
95 % half-width	**0.56 s**
Half-width / estimate	**3.9 %** ✓ ≤ 5 %
95 % CI	[13.97, 15.09] s

The ≤ 5 % precision target is satisfied by this method.

4.2 Method (b): Batch Means on a Single Long Run

One 30 h long run, warmup deleted. Smallest B with $|\hat{\rho}_1| \le 0.10$ was found by search.

Parameter	Value
B (batch size)	**60 completions**
K (batches)	**149**

Parameter	Value
Lag-1 autocorrelation $\hat{\rho}_1$	**0.044** ✓ ≤ 0.10
Point estimate	**≈ 16.6 s**
95 % half-width	**≈ 2.3 s**
95 % CI	≈ [14.3, 18.9] s

> **Note:** The batch-means half-width (~14 % of estimate) exceeds 5 %. This is an inherent limitation of a single 30 h run — achieving 5 % batch-means precision would require ~212 h of simulation. The 5 % requirement is satisfied by method (a).

4.3 CI Sanity Check — Overlap

```
Rep  CI : [13.97, 15.09]
BM   CI : [14.27, 18.93]  (approximately)
Overlap : PASS ✓
```

The minor difference in point estimates is consistent with Monte Carlo variance.

---

5. Task 4 — CRN Paired Comparison

5.1 Methodology

Both policies were run with **identical (`master_seed`, `rep_index`) pairs** (R = 40), using the `seeds.make_streams` CRN mechanism. For each replication r:

$$D_r = \bar{X}_{A,r} - \bar{X}_{B,r}$$

A paired *t*-CI was built from the R differences.

5.2 Results

Quantity	Value
R (replications)	40
Point estimate $\hat{\mu}_D$	**10.41 s**
95 % half-width	**0.44 s**
95 % CI for μ_A − μ_B	**[9.97, 10.85] s**
Half-width / estimate	**4.2 %** ✓ ≤ 5 %
Variance-Reduction Factor (VRF)	**1.75**

5.3 VRF Interpretation

$$\text{VRF} = \frac{\widehat{\text{Var}}(\bar{X}_A) + \widehat{\text{Var}}(\bar{X}_B)}{\widehat{\text{Var}}(\bar{D})} = 1.75$$

CRN reduced the variance of the paired estimator by approximately **43 %** compared to independent sampling, confirming positive correlation was induced across policies.

5.4 Decision

The **entire 95 % CI [9.97, 10.85] s lies strictly above zero** — the null hypothesis $\mu_A = \mu_B$ is rejected at the 5 % level. Policy B reduces mean per-drone swap-queue wait by **≈ 10–11 s** per depot visit.

> **Recommendation: Invest in the sixth swap station (Policy B).**

---

6. Task 5 — Verification & Validation

6.1 V1 — Fleet Conservation Invariant

At every simulated moment: FLEET_SIZE = n_air + n_depot

Check	Result
`state.max_invariant_violation`	**0.0** ✓

No drone was ever lost or double-counted over the entire 30 h replication.

6.2 V2 — Little's Law on the Depot Subsystem

In steady state: $L = \lambda W$

- L_{observed} = $N - \overline{n_{\text{air}}}$ (sampled every 5 s)
- λ = post-warmup completions / post-warmup duration
- W_{depot} = mean per-drone sojourn (swap + test, wait + service)

Quantity	Value
L_{observed}	**6.86 drones**
$L_{\text{predicted}} = \lambda \cdot W_{\text{depot}}$	**6.86 drones**
Ratio	**0.9999** ✓

Within **0.01 %** of 1.0 — well inside the ±5 % tolerance. Both V&V checks pass.

---

7. Summary of Numeric Results

Key	Value
Classification	Steady-state
warmup_welch	8 820 s
warmup_mser	3 329 s
warmup_chosen	**10 000 s**
Policy A mean wait — replications	14.53 s ± 0.56 s
Policy A mean wait — batch means	~16.6 s ± ~2.3 s
A − B paired difference	**10.41 s**
95 % CI (A − B)	**[9.97, 10.85] s**
CRN VRF	1.75
Fleet invariant violation	0.0
Little's Law ratio	0.9999

---

8. Final Recommendation

Purchase the sixth swap station (Policy B). The paired CRN study demonstrates, with 95 % statistical confidence, that the additional station reduces per-drone swap-queue waiting time by 10–11 seconds per depot visit. The improvement is statistically significant (the entire CI excludes zero), consistent across 40 independent paired replications, and backed by a model that passes all V&V checks. The operations manager should proceed with the capital investment.

IE 306 — Systems Simulation, Spring 2026. Files `config.py`, `model.py`, `seeds.py` were not modified. Analysis in `assignment_03_starter.ipynb`.